

Project 1 – Student Management System

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Domain Name: AIML Development

College Name: zeal polytechnic Narhe , Pune

Branch: AIML

Project Code :

```
Mini project.py > calculate_grade
1  # Student Management System
2  students = {}
3  def calculate_grade(per):
4      if per >= 90:
5          return "O"
6      elif per >= 80:
7          return "A+"
8      elif per >= 70:
9          return "A"
10     elif per >= 60:
11         return "B+"
12     elif per >= 50:
13         return "B"
14     else:
15         return "F"
16
17 def add_student():
18     try:
19         roll = int(input("Enter Roll No: "))
20
21         if roll in students:
22             print("Roll number already exists")
23             return
24
25         name = input("Enter Name: ")
26
27         marks = []
28         for i in range(5):
29             marks.append(float(input(f"Enter Subject {i+1} marks: ")))
30
```

```
31     percentage = sum(marks) / 5
32     grade = calculate_grade(percentage)
33
34     students[roll] = {
35         "name": name,
36         "marks": marks,
37         "percentage": percentage,
38         "grade": grade
39     }
40
41     print("Student Added Successfully")
42
43 except ValueError:
44     print("Invalid Input")
45
46
47 def view_all():
48     for roll, data in students.items():
49         print("\nRoll:", roll)
50         print("Name:", data["name"])
51         print("Percentage:", data["percentage"])
52         print("Grade:", data["grade"])
53
54
55 def search_student():
56     roll = int(input("Enter Roll No: "))
57
58     if roll in students:
59         print(students[roll])
60     else:
61         print("Student Not Found")
62
63
64 def update_student():
65     roll = int(input("Enter Roll No: "))
66
```

```

67     if roll in students:
68         name = input("Enter New Name: ")
69         students[roll]["name"] = name
70         print("Updated Successfully")
71     else:
72         print("Not Found")
73
74
75     def delete_student():
76         roll = int(input("Enter Roll No: "))
77
78         if roll in students:
79             del students[roll]
80             print("Deleted Successfully")
81         else:
82             print("Not Found")
83
84
85     while True:
86
87         print("""
88         ===== STUDENT MANAGEMENT SYSTEM =====
89         1.Add Student
90         2.View All
91         3.Search
92         4.Update
93         5.Delete
94         6.Exit
95         """)
96
97         choice = int(input("Enter Choice: "))
98
99         if choice == 1:
100             add_student()
101
102         elif choice == 2:

```

```

102         elif choice == 2:
103             view_all()
104
105         elif choice == 3:
106             search_student()
107
108         elif choice == 4:
109             update_student()
110
111         elif choice == 5:
112             delete_student()
113
114         elif choice == 6:
115             break
116
117         else:
118             print("Wrong Choice")

```

Output:

```
===== STUDENT MANAGEMENT SYSTEM =====
1.Add Student
2.View All
3.Search
4.Update
5.Delete
6.Exit

Enter Choice: 1
Enter Roll No: 14
Enter Name: shruti
Enter Subject 1 marks: 80
Enter Subject 2 marks: 77
Enter Subject 3 marks: 67
Enter Subject 4 marks: 87
Enter Subject 5 marks: 91
Student Added Successfully

===== STUDENT MANAGEMENT SYSTEM =====
1.Add Student
2.View All
3.Search
4.Update
5.Delete
6.Exit

Enter Choice: 2

Roll: 14
Name: shruti
Percentage: 80.4
Grade: A+

===== STUDENT MANAGEMENT SYSTEM =====
1.Add Student
2.View All
3.Search
4.Update
5.Delete
6.Exit

Enter Choice: 3
Enter Roll No: 25
Student Not Found
```

```
===== STUDENT MANAGEMENT SYSTEM =====
1.Add Student
2.View All
3.Search
4.Update
5.Delete
6.Exit

Enter Choice: 4
Enter Roll No: 14
Enter New Name: shruti
Updated Successfully

===== STUDENT MANAGEMENT SYSTEM =====
1.Add Student
2.View All
3.Search
4.Update
5.Delete
6.Exit

Enter Choice: 5
Enter Roll No: 16
Not Found

===== STUDENT MANAGEMENT SYSTEM =====
1.Add Student
2.View All
3.Search
4.Update
5.Delete
6.Exit

Enter Choice: 6
PS C:\Users\tanve\Desktop\ITR> 
```